ELECTRICITY -11

Class 10 - Science

Section A

1. **State True or False:** **[19]**

(a) A conductor having some appreciable resistance is called a resistor. **[1]**

(b) A continuous and closed path of an electric current is called an electric circuit. **[1]**

(c) In circuits using metallic wires, electrons constitute the flow of charges. **[1]**

(d) One ampere is constituted by the flow of one coulomb of charge per second. **[1]**

(e) Tungsten is used almost exclusively for filaments of electric bulbs. **[1]**

(f) When current is passed through a high resistance wire, the wire becomes very hot and produces heat, this effect is known as the heating effect of current. **[1]**

(g) A switch makes a conducting link between the cell and the bulb. **[1]**

(h) The electric potential difference between two points in an electric circuit, carrying some current, is the amount of work done to move a unit charge from one point to another. **[1]**

(i) In an electric circuit, a device called rheostat is often used to change the resistance in the circuit. **[1]**

(j) Copper and aluminium are generally used for electrical transmission lines. **[1]**

(k) Voltmeter is a device used for the measurement of current. **[1]**

(l) The potential difference is measured by means of an instrument called the galvanometer. **[1]**

(m) A drawing showing the way various electric devices are connected in a circuit is called a circuit diagram. **[1]**

(n) The ammeter is always connected in parallel across the points between which the potential difference is to be measured. **[1]**

(o) The motion of electrons in an electric circuit constitutes an electric current. **[1]**

(p) The property of the conduct due to which it opposes the flow of current through it is called resistance of the conductor. **[1]**

(q) When an electrical appliance consumes electrical energy at the rate of 1 joule per second, its power is said to be 1 watt. **[1]**

(r) A series circuit divides the current through the electrical gadgets. **[1]**

(s) A component used to regulate current without changing the voltage source is called variable resistance. **[1]**

Section B

2. **Fill in the blanks:** **[20]**

(a) When current is passed through a high resistance wire, the wire becomes very hot and produces heat, this effect is known as ________ of current. **[1]**

(b) The conventional direction of electric current is the one in which ________ charges move orderly. **[1]**

(c) The commercial unit of electric energy is ________. **[1]**

(d) A continuous and closed path of an electric current is called an ________. **[1]**

(e) When an electrical appliance consumes electrical energy at the rate of 1 joule per second, its power is said to be ________. **[1]**

(f) ________ is used almost exclusively for filaments of electric bulbs. **[1]**

(g) The S.I. unit of ________ is ohm metre. **[1]**

(h) In an electric circuit, a device called ________ is often used to change the resistance in the circuit. **[1]**

(i) A ________ showing the way various electric devices are connected in a circuit is called a circuit diagram. **[1]**

(j) ________ is the rate at which electrical energy is produced or consumed in an electric circuit. **[1]**

(k) Those substances through which electricity cannot flow are known as ________. **[1]**

(l) The potential difference is measured by means of an instrument called the ________. **[1]**

(m) A component used to regulate the current without changing the voltage source is called ________. **[1]**

(n) A ________ circuit divides the current through the electrical gadgets. **[1]**

(o) A conductor having some appreciable resistance is called a ________. **[1]**

(p) ________ is a device used for the measurement of current. **[1]**

(q) A component of identical size that offers a ________ resistance is a poor conductor. **[1]**

(r) ________ states that at a constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends. **[1]**

(s) The ________ at a point in a field is defined as the work done in moving a unit charge from infinity to that point. **[1]**

(t) One ________ is constituted by the flow of one coulomb of charge per second. **[1]**

Section C

3. Which of the given statement is true or false: **[1]**

Statement A: Resistivity increases with a decrease in temperature in insulators.

Statement B: Resistivity of a conductor increases with increasing temperature.

a) Statement A is true, Statement B is false. b) Neither statement A nor statement B is true.

c) Statement A is false, Statement B is true. d) Both the statements A and B are true.

4. Which of the given statement is true or false: **[1]**

Statement A: Light from a bathroom bulb gets dimmer for a moment when the geyser is switched on.

Statement B: Insulators conduct charges, they can be charged easily by friction.

a) Both the statements A and B are true. b) Neither statement A nor statement B is true.

c) Statement A is true; Statement B is false. d) Statement B is true; Statement A is false.

5. A student performing an experiment with a series combination of two resistors infers that **[1]**

A: Potential difference across the combination is the sum of the potential difference across each of them.

B: The current passing through them is the same.

Which of the following statement is/are true?

a) Both A and B are false b) B is true and A is false

c) A is true and B is false d) Both A and B are true

6. When two objects are rubbed together, one object loses some of its free electrons while the other gains the same **[1]**

number of these free electrons. Which object gains the positive charge?

A. The object losing electrons.

B. The object gaining electrons.

a) Both (A) and (B)

b) (B)

c) None of these

d) (A)

Section D

7. Which of the following statements is/are incorrect? **[1]**

A. A neutron has a positive charge of $+1.6 \times 10^{-19}$ C

B. An ammeter is a low resistance device.

C. The resistance of semiconductors decreases with temperature.

D. One ampere is equal to 10^{-6} uA.

a) A and C

b) A and D

c) A and B

d) A, B and C

8. Find the incorrect statement: **[1]**

A. The direction of flow of negative charge was taken to be the direction of the electric current.

B. The P. D. for all the resistors in the parallel circuit remains constant.

C. The SI unit of electric charge is the coulomb.

a) None of these

b) (A)

c) (B)

d) (C)

9. Which of the following is incorrect about the heat produced in a resistor? **[1]**

A. It is directly proportional to the square of the current.

B. Directly proportional to the resistance for a given current.

C. Directly proportional to the time for which the current flows through the resistors.

a) None of these

b) (B)

c) (C)

d) (A)

10. A student says that the resistance of two wires of the same length and same area of cross-section is the same. This statement is correct if**:** **[1]**

a) Both wires are made of same material and are at same temperature.

b) Both wires are made of different materials and are at the same temperature.

c) Both wires are made of same material and are at different temperature.

d) Both wires are of different materials.

11. Which of the following statements are correct about resistance? **[1]**

A. The resistance of a conductor is directly proportional to the length.

B. The resistance of a conductor is directly proportional to the area of cross-section.

C. The Resistance of a conductor depends on the nature of the material.

D. The Resistance of a conductor is inversely proportional to the length.

a) A, B and D

b) A and B

c) A and C

d) A and D

12. Which statement/statements is/are correct? **[1]**

a) A voltage has a low resistance

b) An ammeter is connected in series in a circuit and a voltmeter is connected in parallel

c) An ammeter has a high resistance

d) None of these

Section E

13. A student uses as battery of adjustable voltage 0-6 V. she has to perform an experiment to determine the equivalent resistance of two resistors when connected in parallel using two resistors of value 3 ohm and 6 ohm. The best choice of combination of voltmeter and ammeter to be used in the experiment is **[1]**

a) Ammeter of range 0-5 A and voltmeter of range 0-2 V.

b) Ammeter of range 0-2 A and voltmeter of range 0-10 V.

c) Ammeter of range 0-5 A and voltmeter of range 0-5 V.

d) Ammeter of range 0-5 A and voltmeter of range 0-10 V

14. The following apparatus is available in the laboratory **[1]**

Battery: adjustable from 0 to 6 V

Resistors: 4Ω and 12Ω

Ammeters: A_1 and Range 0 to 5 A; Least Count 0.25 A: A_2 and Range 0 to 3 A; Least Count 0.1 A

Voltmeters: V_1 of Range 0 to 10 V; Least Count 0.5 V: V_1 of Range 0 to 5 V; Least Count 0.1 V

For the experiments to find the equivalent resistance of the parallel combination of the two given resistors, the best choice would be:

a) ammeter A_2 and voltmeter V_1

b) ammeter A_2 and voltmeter V_2

c) ammeter A_1 and voltmeter V_2

d) ammeter A_1 and voltmeter V_1

15. In an experiment on finding the equivalent resistance of two resistors, connected in series, a student connects the terminals of the voltmeter, to **[1]**

a) both the terminals of each of the two resistors.

b) both the terminals of one resistor and one terminal of the other resistor.

c) one terminal of each of the two resistors and these terminals are also interconnected.

d) one terminal of each of the two resistors and these terminals are not interconnected.

16. The following precautions were listed by a student in the experiment on study of Dependence of current on potential difference. **[1]**

A. Use copper wires as thin as possible for making connections.
B. All the connections should be kept tight.
C. The positive and negative terminals of the voltmeter and the ammeter should be correctly connected.
D. The zero error in the ammeter and the voltmeter and the voltmeter should be noted and taken into consideration while recording the measurements.
E. The key in the circuit, once plugged in, should not be taken out till all the observations have been completed.

The precautions that need to be corrected and revised are:

a) B and E
b) A, C and E
c) A and E
d) C and E

17. A student did the experiment to find the equivalent resistance, of two given resistors, R_1 and R_2, first when they are connected in series and next when they are connected in parallel. The two values of the equivalent resistance obtained by him were R_S and R_P respectively. He would find that **[1]**

a) $R_S > R_P$
b) $R_P > R_S$
c) $R_S = R_P$ but not equal to $\frac{R_1+R_2}{2}$
d) Rs = Rp = $\frac{R_1+R_2}{2}$

18. The following apparatus is available in a laboratory. **[1]**

Cell: Adjustable from 0 to 1.5 volt

Resistor: 4Ω and 12 Ω

Ammeters: A_1 of Range 0 to 3 A; Least Count 0.01 A : A_2 of Range 0 to 1 A; Least Count 0.05 A

Voltmeters: V_1 of Range 0 to 10 V; Least Count 0.5 V: V_2 of Range 0 to 5 V; Least Count 0.1 V

The best combination of voltmeter and ammeter for finding the equivalent resistance of the resistors in parallel would be

a) ammeter A_1 and voltmeter V_1
b) ammeter A_2 and voltmeter V_2
c) ammeter A_2 and voltmeter V_1
d) ammeter A_1 and voltmeter V_2

19. Three students X, Y and Z, while performing the experiment to study the dependence of current on the potential difference across a resistor, connect the ammeter (A), the battery (B), the key (K) and the resistor (R), in series, in the following three different orders. **[1]**

X $\longrightarrow$ B, K, R, A, B

Y $\longrightarrow$ B, A, K, R, B

Z $\longrightarrow$ B, R, K, A, B

Who has connected them in the correct order?

a) Z
b) X
c) X, Y and Z
d) Y

20. The following instruments are available in a laboratory :Milliammeter A_1 of range 0-300 mA and least count 10 mAMilliammeter A_2 of range 0-200 mA and least count 20 mAVoltmeter V_1 of range 0-5 V and least count 0.2 VVoltmeter V_2 of range 0-3 V nd least count 0.3 V. **[1]**

Out of the following pairs of instruments, which pair would be the best choice for carrying out the experiment to determine the equivalent resistance of two resistors connected in series ?

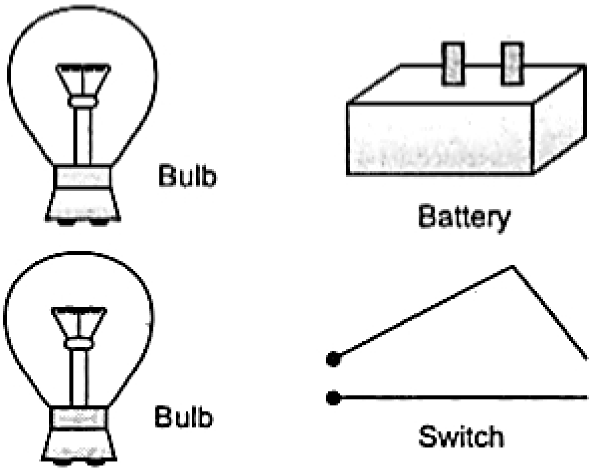
a) Milliammeter A_2 and voltmeter V_2
b) Milliammeter A_1 and voltmeter V_1
c) Milliammeter A_2 and voltmeter V_1
d) Milliammeter A_1 and voltmeter V_2

Section F

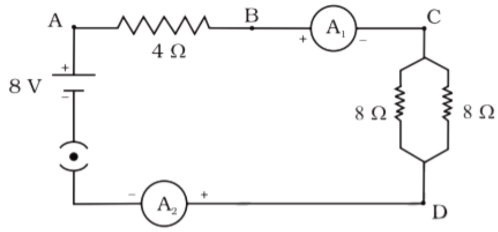
21. **[3]**
a. What is the heating effect of electric current?
b. Write an expression for the amount of heat produced in a resistor when an electric current is passed through it stating the meanings of the symbols used.
c. Name two appliances based on heating effect of electric current.

22. Judge the equivalent resistance when the following are connected in parallel (a) 1Ω and $10^6\ \Omega$ (b) 1Ω and $10^3\Omega$ and $10^6\Omega$ **[3]**

23. The given figure shows a battery, a switch and two bulbs. Complete the diagram to show the electric connections of the bulbs to the battery. How have you joined the bulbs? Give a reason. **[3]**



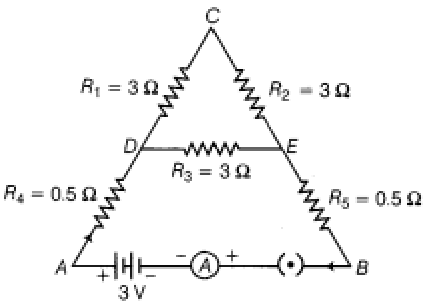
24. Find out the following in the electric circuit given in Figure **[3]**



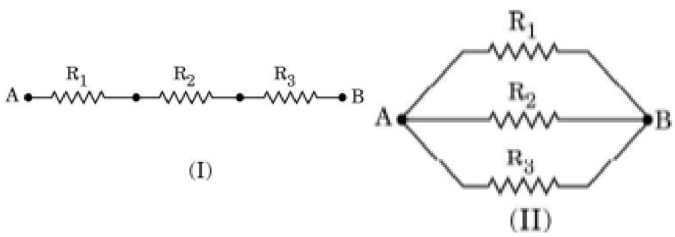
i. The potential difference across 4 Ω resistance
ii. The power dissipated in 4 Ω resistor

25. A metallic wire of resistance R is cut into ten parts of equal length. Two pieces each are joined in series and then five such combinations are joined in parallel. What will be the effective resistance of the combination? **[3]**

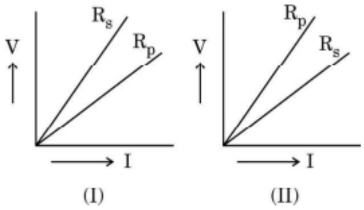
26. Five resistors are connected in a circuit as shown in figure. Find the ammeter reading when the circuit is closed. **[3]**



27. i. Write the formula for determining the equivalent resistance between A and B of the two combinations (I) and (II) of three resistors R_1, R_2 and R_3 arranged as follows: **[3]**



ii. If the equivalent resistance of the arrangements (I) and (II) are R_s and R_p respectively, then which one of the following VI graphs is correctly labelled? Justify your answer.

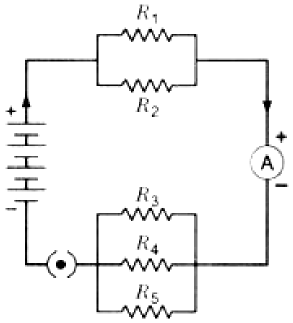


28. Calculate the total cost of running the following electrical devices in the month of September, if the rate of 1 unit of electricity is Rs. 6.00. **[3]**

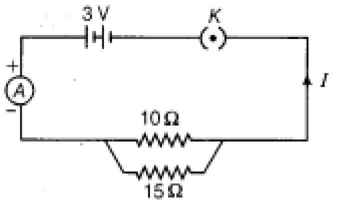
   i. Electric heater of 1000 W for 5 hours daily.
   ii. Electric refrigerator of 400 W for 10 hours daily.

29. If in the figure R_1= 10Ω, R_2 = 40Ω, R_3 = 30Ω, R_4 = 20Ω, R_5 = 60Ω, and a 12 V battery is connected to the arrangement.Calculate **[3]**

   i. the total resistance in the circuit ,and
   ii. the total current flowing in the circuit.



30. Study the following circuit and answer the questions that follows: **[3]**



   i. How much current is flowing through
      a. 10Ω and
      b. 15Ω resistor?
   ii. What is the ammeter reading?

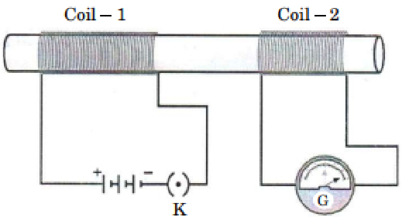
31. a. List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. **[3]**
    b. The resistance of a wire of 0.01 cm radius is 10Ω. If the resistivity of the wire is 50×10^{-8} Ωm, find the length of this wire.

32. Show how would you join three resistors, each of resistance 9 Ω so that the equivalent resistance of the combination is **[3]**

   i. 13.5 Ω
   ii. 6 Ω?

33. i. When a 12 V battery is connected across an unknown resistor, there is a current of 2.5 mA in the circuit. Find the value of the resistance of the resistor? **[3]**
    ii. 320J of heat is produced in 10 s in a 2Ω resistor. Find the amount of current flowing through the resistor.

34. In the diagram given below, Coil - 1 is connected in series with a battery and a plug key while Coil - 2 is connected with a galvanometer. **[3]**



i. Why does the galvanometer show deflection only when the key (K) is plugged in and not when a steady current starts flowing in the circuit?
ii. What is observed in the galvanometer, when the key is plugged out?
iii. State the conclusion based on the observation of this activity.

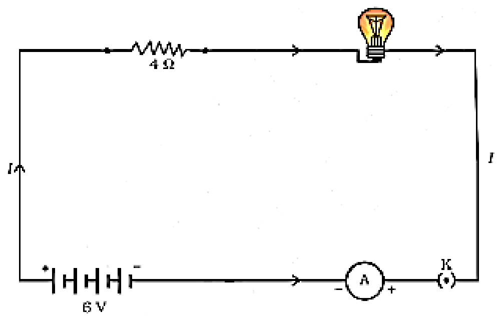
35. Three students X, Y and Z while performing the experiment to study the dependence of current on the potential difference across a resistor, connect the ammeter (A), the battery (B), the key (k) and the resistor (R) in series, in the following three different orders. **[3]**

i. X$\rightarrow$ B, K, R, A, B
ii. Y$\rightarrow$ B, A, K, R, B
iii. Z$\rightarrow$B, R, K, A, B

Who has connected them in the correct order?

36. i. An electric iron consumes energy at a rate of 840 W when heating is at the maximum rate and 360 W when the heating is at the minimum rate. The applied voltage is 220 V. What is the value of current and the resistance in each case? **[3]**
ii. Which uses more energy, a 250 W TV set for 1 hour or a 1,200 W toaster for 10 minutes?

37. An electric lamp, whose resistance is 20Ω, and a conductor of 4Ω resistance are connected to a 6 V battery in figure. **[3]**
Calculate (a) the total resistance of the circuit, (b) the current through the circuit, and (c) the potential difference across the electric lamp and conductor.



38. i. A hot plate of an electric oven connected to a 220 V line has two resistance coils A and B, each of 24Ω resistance, which may be used separately, in series, or in parallel. What are the currents in the three cases? **[3]**
ii. Calculate the resistance of an electric bulb that allows a 10A current when connected to a 220V power source?